

FAIR-D: An Extension to the FAIR Principles for Open Hardware Replicability in Human-Computer Interaction

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Abstract

Replicability is a cornerstone of scientific progress, yet in Human-Computer Interaction (HCI), research involving novel hardware often faces significant barriers to replication. Prototypes can be expensive, rely on specialized components, or require complex assembly, hindering validation and adoption by the wider community. This paper argues for an "Open Science for HCI" approach, advocating for the design of prototypes that are intentionally simple to build and repair, and are founded on the most accessible technological solutions available. We present this framework and demonstrate its practical application via current SPART (On-Surface Positioning Augmented Reality) prototypes. SPART is a peephole interaction that recreates the collaborative affordances of interactive tabletops with standard mobile devices. Insights from a usability study conducted with 67 visitors at a public science fair illustrate SPART's position in the framework and support our discussion on how this framework can lead to a more robust, collaborative, and innovative HCI research ecosystem.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Haptic devices; Ubiquitous and mobile computing; Ubiquitous and mobile computing theory, concepts and paradigms; Mobile computing; Accessibility; Accessibility theory, concepts and paradigms.

Keywords

Open Science, Reproducibility, FAIR, Open Hardware, HCI, Prototyping, Augmented Reality, Collaborative Learning, Interactive Tabletops, DIY

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1 Introduction

Human-Computer Interaction (HCI) is a fundamentally constructive field in which interfaces and systems are designed to explore new forms of interaction 1. However, research prototypes often face issues of replicability and generalizability: one-of-a-kind or complex systems are costly to reproduce 2, contributing to the broader "replication crisis" 3 and a focus on small-scale validation [4, 5]. Multi-laboratory experiments 6 and citizen science [7, 8] offer a path toward more horizontal research, promising scalable participation, improved replicability, and continuous design feedback 9. Related ideas appear in agile and PI-DBR approaches [10, 11], which extend the "accessible" dimension of the FAIR principles toward broad usability 12.

Yet, accessibility and sharing remain limited in HCI, especially for hardware: only about one-third of artifacts are shared, often without sufficient documentation 13. While horizontality can mitigate risks such as design fixation and groupthink [14–16] by involving external communities 17, open science in HCI remains difficult because artifacts combine software and hardware. Sharing alone is insufficient; missing dependencies and documentation often block replication 13, and hardware further requires detailed fabrication and assembly documentation, as reflected in recent adaptations of FAIR for open hardware 12, 18, 19]. Experience from software ecosystems shows that documentation debt persists even there [20, 21].

We therefore propose FAIR-D (Design) principles to integrate open science through accessible, well-documented data, software, and hardware into the design process, and demonstrate them through the SPART project, which provides tabletop-like collaborative affordances without their cost and immobility. This paper contributes the FAIR-D framework extension, a case study of a FAIR-D compliant design (SPART-ME), and a real-world validation of its learnability and robustness.

2 Contributions

2.1 FAIR-D - An Extended Open Science Design Framework for Horizontal Research

Extending on recent tendencies toward multi-laboratory experiments as a manifestation of horizontality in science, a completely horizontal, theoretical research ecosystem can be described along dimensions of skills, resources, and communication needs of its actors. This ecosystem spans from the core research team (1), which communicates through direct and immediate collaboration, to the broader HCI community (2), where interaction occur primarily through publications, conferences, and occasional direct exchanges.

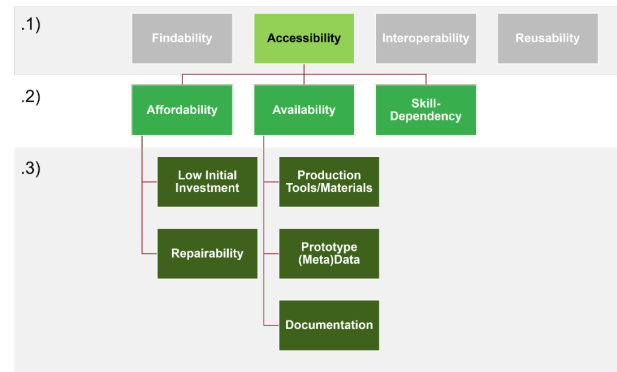


Figure 1: FAIR-D principles (grey: unchanged FAIR principles, green: extended accessibility principle)

It further extends to a peripheral field of tinkerer and hacker communities or other domain stakeholders (3), which engages through scientific media, community forums, and public events, and can actively innovate on shared prototypes. Finally, it includes end users (4), who typically have limited technical expertise and interact with the artifacts through general media or organic dissemination channels, but whose participation is essential for large-scale adoption and evaluation.

From the above classification, we deduce artefact attributes depending on the horizontality envisioned. The expected decline in technical and research skills towards peripheral actors results, for instance, in a need for an increasingly simplified process to obtain and operate a research prototype. We thus propose the following extension of the FAIR accessibility principles (Figure 1.): More about the submission template

- **Affordability** is defined through the initial investment required for prototype parts and overall reparability to reduce follow-up costs (see Figure 1.2).
- A **low initial investment** to make the barrier to engaging with the project as low as possible, both for research institutions and end users (Figure 1.3).
- **Reparability** ensures that the initial investment is not substantially surpassed (Figure 1.3).
- **Availability** concerns the availability of production tools, availability of prototype data (e.g., 3D files), and documentation to the user group (Figure 1.2).
- **Documentation** must be tailored to different actors (Figure 1.3): communities need technical build and extension documentation, end users and testers need simple usage guides, while researchers require detailed experimental protocols for exact replicability.
- Shared **prototype (meta)data** must be adapted to levels of peripherality (Figure 1.3): while researchers and maker communities benefit from source code and native design files, end users need packaged, ready-to-use versions, and data collection should be either guided or embedded in the system, while still preserving FAIR’s original accessibility principle.
- **Production tools** and materials should be widely available (Figure 1.3), as reliance on specialized facilities or hardware,

both for fabrication and software compilation reduces accessibility, while commonly available tools in places like makerspaces ensure broad reproducibility.

- **Skill-dependency** means the required skills that the different user groups require to set up and operate the research prototype (Figure 1.2).

2.2 SPART - A Case Study of Open Science

Since the pioneering work on systems like the CoLab at Xerox PARC 22, HCI has extensively explored interactive tabletops as a key paradigm for supporting co-located collaboration 23, demonstrating clear benefits for group work and participation 24. However, due to their high cost, immobility, and infrastructural demands, such systems remain largely confined to labs and public installations, rarely seeing widespread adoption in everyday contexts such as classrooms or meeting rooms. Mobile devices offer an affordable and portable alternative for collaborative learning, as explored in mCSCL and split-screen approaches 25, but their limited screen size remains a key constraint. While AR can overcome this limitation, handheld AR induces physical fatigue and poor collaborative dynamics (“gorilla arm”) 26, and head-mounted displays introduce issues of cost, weight, and discomfort 27. This creates a clear design gap for solutions that combine the collaborative affordances of tabletops with the affordability and portability of mobile devices.

SPART implements the “magic lens” or “peephole” metaphor 28, specifically its dynamic variant where users explore a large shared digital workspace by physically moving a mobile device across a surface 29. Similar to systems such as HuddleLamp 30 and JuxtaPinch 31, it supports spatially aware, multi-device collaboration, but distinguishes itself through a low-cost, accessible, and easily reproducible implementation of the 2D dynamic peephole interaction.

2.3 SPART - Usability Study & Implications of Prototyping for FAIR-D

FAIR-D principles have guided SPART since its first proofs of concept (POC) to enable horizontal development beyond the core research team. These principles were operationalized into concrete criteria covering findability, accessibility (affordability <€ 50, reparability, and availability through common tools, public repositories,

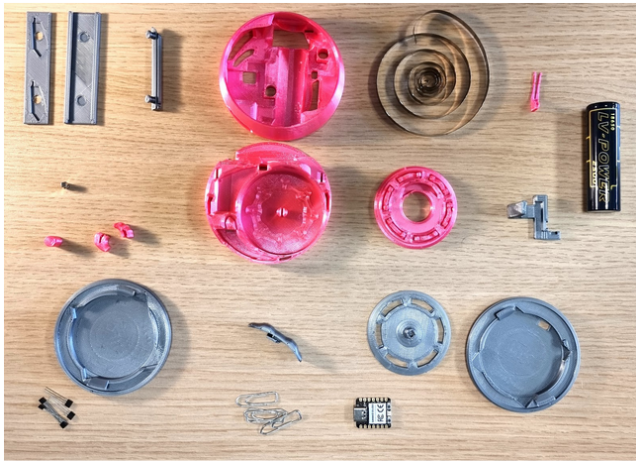


Figure 2: SPART-ME M required parts (left) and SPART-ME M assembled prototype (right)

and documentation), low skill requirements, interoperability with standard mobile devices, and reusability through modular design. In addition, the system had to meet core interaction requirements, including high accuracy (≤ 1 cm), sufficient fluidity (≥ 10 Hz), A3-scale coverage, portability, robustness, and support for at least two devices.

The core research team iteratively evaluated multiple tracking technologies against these functional FAIR-D criteria. Magnetic (SPART-MA), acoustic (SPART-AC), and vision-based approaches were discarded due to insufficient robustness, accuracy, or accessibility, leading to mechanical tracking as the most promising avenue. An initial dual-string prototype (SPART-ME D) met performance requirements but exceeded the affordability target, relying on low-torque 10 turn potentiometers (each 20 €) and a dedicated analog to digital converter chip ($\sim$ € 60). The development of a low-cost, printable magnetic rotary encoder ($\sim$ € 2 in parts, 08/2025) based on widely available analog SS49 Hall sensors, enabled a compact, single-string design. The current variant relies on commonly available components: a PLA printed mechanism and casing, two 3mm cube N42 neodymium magnets an NRF52480 XIAO Chip, four SS49 analog Hall sensors, A 18650 Lithium battery, a spiral spring from a retractable card holder and paperclips (see Figure 2). SPART-ME M communicates exposes a Bluetooth GATT (ensuring interoperability), supports modular sensor extensions (e.g., NFC via I2C), and is housed in a glue-free, fully repairable 3D-printed case, fulfilling reusability and reparability goals. 3D files, parts list, bill of materials, assembly instructions and documentation are publicly available on the project’s OSF repository.

Usability and learnability were validated in an in-situ study with 67 participants at the “Le Mans Sonore” science fair using the SPART-ME M prototype. Interaction logs showed high engagement (mean session time of 3:50 min) and strong adoption of the intended interaction (3,478 SPART movements vs. 180 touch interactions), while erroneous gestures were negligible ($<0.1\%$) and decreased exponentially with use. While this demonstrates the benefits of accessible, low-cost, and robust prototypes, even for unsupervised

public use, it also highlights the need to involve external actors earlier: although firmware and 3D files are public, the mobile software still carries technical debt and lacks sufficient documentation.

3 Discussion and Conclusion

This paper introduced the FAIR-D framework as a design-oriented extension of the FAIR principles for HCI prototypes, grounded in a horizontal research perspective. The FAIR-D extends existing practices mostly found in software development to the realm of hardware development of research prototypes, inciting designers to consider replicability and validation through accessibility early on in the design process. It supports this reflection through the consideration of the different potential user types across peripherality. By adhering to this design thinking process, HCI research prototypes are positioned not only as demonstrators of new human-computer interactions, but also as an artefact to foster inter-community engagement and collaboration. Through the SPART project, we demonstrate that it is possible to build high-performance HCI prototypes that are simultaneously open, accessible, and affordable, thereby lowering barriers for replicability and collaborative research. The development of SPART-ME M shows that prioritizing affordability, DIY-friendliness, and reparability from the outset can guide design toward solutions that are both effective and easily replicable, without compromising innovation or functionality.

We argue that replicability must be considered early in the design of HCI prototypes, particularly for hardware-based systems. Current work focuses on disseminating SPART-ME M to maker communities and educators, where its simple fabrication process (requiring little more than a soldering iron and a 3D printer) also enables educational use and further extension by students. More broadly, SPART illustrates how extending FAIR principles to both software and hardware can help address the replicability challenge in HCI. While not all prototypes will reach this level of accessibility, a sustained community commitment to FAIR-D principles is, in our view, essential for a more open, cumulative, and collaborative HCI research practice.

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